Andrew J. Peng

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EDUCATION

Bachelor of Science in Computer Engineering — Expected Fall 2028
University of Maryland — College Park, MD

- **GPA**: 3.96 | A. James Clark School of Engineering Dean's List: F24, S25, F25, S26

EXPERIENCE

Machine Learning Research Assistant — May 2026 – Aug. 2026
Shanghai Jiao Tong University — Shanghai, China

- Conducted **few-shot object detection** research for satellite imagery on a **4,481-image dataset** spanning **25 classes** of ships, military aircraft, and launch vehicles, supervised by **Prof. Hongtian Chen**, collaborating with graduate researchers.
- Benchmarked **RF-DETR Small** using **COCO-style** whole-image evaluation, **worst-image ranking**, and **SAHI/tiled inference** comparisons to identify the most reliable detection strategy for **small, few-shot** classes.
- Diagnosed model accuracy errors through per-class and confusion-matrix analysis, uncovering **dataset labeling inconsistencies**; used **DarkLabel** to manually review and correct mislabeled annotations.

Game Developer — Dec. 2024 – Present
Contractor — Belmont, CA

- Utilized **Luau** to design and deliver **30+ modular game systems**, including data management, UI/UX, gameplay mechanics, and physics simulations for client commissions on **Roblox**.
- Contracted for Tiny Lab Studio; designed the combat framework and exploit prevention, including rate limiting and server-side validation, for **Push Party**, which has drawn **50,000+ visits**, **20,000+ new players**, and up to **100 concurrent players** since launch; continue supporting the game through bug fixes and QA testing.

Vehicle Dynamics & Ergonomics Engineer — Aug. 2024 – May 2026
Terps Racing — College Park, MD

- Designed **ABS housings** using **SolidWorks** and integrated **brake temperature sensors** and **push rod strain gauges** for the 2026 FSAE EV; conducted FEA on sensor mounts to validate structural integrity against vibrational racing loads.
- Modeled a **clutch actuation** assembly using **SolidWorks** for the 2025 FSAE IC vehicle; conducted FEA on the Onyx carbon lever and 6061 aluminum mount to validate structural integrity under **150 N peak input** and vibrational loads; contributed to vehicle manufacturing via **water jet machining** and **3D printing**.

PROJECTS

LiteLabel — Sept. 2026 – Oct. 2026

- Built a **bounding-box image annotation tool** for object detection datasets, exporting **YOLO**, **RF-DETR**, **Pascal VOC**, and custom text or CSV label formats from a **YAML** config file.
- Shipped standalone **Windows** and **macOS** releases built automatically with **GitHub Actions**, with hotkey navigation, auto-save, and a **pytest** suite.

ProfileLite — Dec. 2025 – Jan. 2026

- Wrote a **type-safe** Roblox DataStore wrapper providing **session-locking**, **auto-saving**, and **cross-server conflict resolution**, using a **ProfileService**-style API; published on the Roblox Creator Store and shared with other developers.
- Implemented an automatic **mock-store** fallback for API-disabled Studio development and paginated **version-history** querying, utilizing a **custom**, **yield-safe** signal system for reliability diagnostics.

SKILLS

Tools: Git, Apache Subversion, LM Studio, DarkLabel, YOLOv8, RF-DETR, Arduino, Raspberry Pi, SolidWorks
Languages: Python, Luau, Java, MATLAB, HTML/CSS
Frameworks: PyTorch, OpenCV, NumPy, Pandas, Matplotlib